

GED Algebra Practice: Algebra Basics, Expressions, and Polynomials

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This worksheet contains 20 practice problems aligned to GED Algebra fundamentals. **Do not** use a calculator unless instructed by your teacher. Show all work.

Part A: Problems

Work each problem. Full solutions appear on page 4.

1. Solve for x : $x - 9 = 14$.
2. Solve for x : $5x + 7 = 2$.
3. Solve for x : $\frac{x}{6} - 3 = 1$.
4. Solve for x : $-4(2 - x) = 12$.
5. A number decreased by 15 is equal to twice the number minus 27. Find the number.
6. The perimeter of a rectangle is 54 cm. The length is 3 cm more than twice the width. Find the width.
7. Solve the system by reasoning: If 3 more than a number is 20, what is twice the number?
8. Simplify completely: $7x - 3 + 5x + 8$.
9. Simplify: $2(3x - 5) - (x + 7)$.
10. Simplify: $6 - [2 - (3x - 4)]$.
11. Evaluate $4y - 2(3y - 5)$ when $y = -2$.
12. Given $a = 3$ and $b = -5$, compute $2a^2 - ab + b$.
13. Write an equivalent expression by factoring: $12x + 18$.
14. Add: $(3x^2 - 2x + 5) + (x^2 + 7x - 3)$.
15. Subtract: $(5x^2 + 6x - 4) - (2x^2 - x + 9)$.
16. Multiply: $-3x(4x - 7)$.
17. Expand: $(x + 6)(x - 2)$.

18. Expand: $(2x - 3)(x + 5)$.
19. Factor completely: $x^2 + 9x + 14$.
20. Factor completely: $6x^2 + 11x + 3$.

Part B: Full Solutions

1. $x - 9 = 14$
Add 9 to both sides: $x = 23$.
2. $5x + 7 = 2$
Subtract 7: $5x = -5$. Divide by 5: $x = -1$.
3. $\frac{x}{6} - 3 = 1$
Add 3: $\frac{x}{6} = 4$. Multiply by 6: $x = 24$.
4. $-4(2 - x) = 12$
Distribute: $-8 + 4x = 12$. Add 8: $4x = 20$. Divide by 4: $x = 5$.
5. Let the number be n . "A number decreased by 15" is $n - 15$. Twice the number minus 27 is $2n - 27$. Set equal: $n - 15 = 2n - 27$. Subtract n : $-15 = n - 27$. Add 27: $n = 12$.
6. Perimeter $P = 54$, width w , length $\ell = 2w + 3$. Use $P = 2(\ell + w)$:
 $54 = 2((2w + 3) + w) = 2(3w + 3) = 6w + 6$.
Subtract 6: $48 = 6w$. So $w = 8$ cm. (Length = 19 cm.)
7. "3 more than a number is 20" $\Rightarrow n + 3 = 20$, so $n = 17$. Twice the number = $2n = 34$.
8. $7x - 3 + 5x + 8 = (7x + 5x) + (-3 + 8) = 12x + 5$.
9. $2(3x - 5) - (x + 7) = 6x - 10 - x - 7 = 5x - 17$.
10. $6 - [2 - (3x - 4)] = 6 - [2 - 3x + 4] = 6 - [6 - 3x] = 6 - 6 + 3x = 3x$.
11. Substitute $y = -2$: $4(-2) - 2(3(-2) - 5) = -8 - 2(-6 - 5) = -8 - 2(-11) = -8 + 22 = 14$.
12. $a = 3$, $b = -5$: $2a^2 - ab + b = 2(9) - (3)(-5) + (-5) = 18 + 15 - 5 = 28$.
13. Factor the GCF: $12x + 18 = 6(2x + 3)$.
14. $(3x^2 - 2x + 5) + (x^2 + 7x - 3) = 4x^2 + 5x + 2$.
15. $(5x^2 + 6x - 4) - (2x^2 - x + 9) = (5x^2 - 2x^2) + (6x - (-x)) + (-4 - 9) = 3x^2 + 7x - 13$.
16. $-3x(4x - 7) = -12x^2 + 21x$.
17. $(x + 6)(x - 2) = x^2 - 2x + 6x - 12 = x^2 + 4x - 12$.
18. $(2x - 3)(x + 5) = 2x^2 + 10x - 3x - 15 = 2x^2 + 7x - 15$.
19. $x^2 + 9x + 14$. Find numbers that multiply to 14 and add to 9: 2 and 7. So $(x + 2)(x + 7)$.
20. $6x^2 + 11x + 3$. Factor by grouping:
Find two numbers with product $6 \cdot 3 = 18$ and sum 11: 9 and 2.
Rewrite middle term: $6x^2 + 9x + 2x + 3$.
Group: $(6x^2 + 9x) + (2x + 3) = 3x(2x + 3) + 1(2x + 3) = (2x + 3)(3x + 1)$.

Teacher notes: Items 1–7 target one- and two-step equations and simple modeling; 8–13 emphasize structure and evaluation; 14–20 address polynomial operations and factoring.